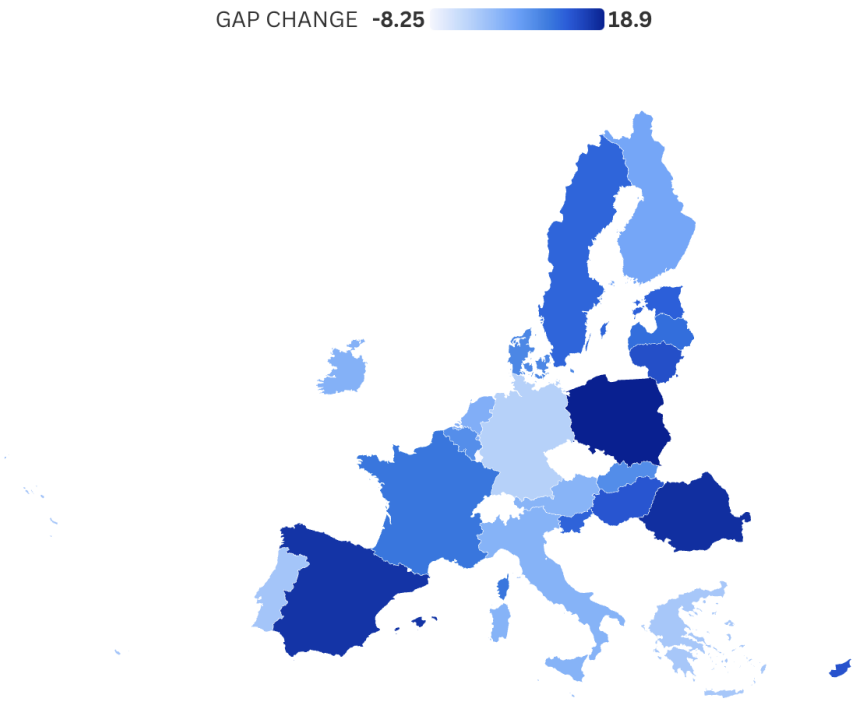
Data Visualization Pitch

Student Individual Assignment



Source: World Bank (boundaries), Simple maps (points)

Bridging the Gap? Urban-Rural Internet Convergence in the EU (2006-2025)

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Research questions

01

- IS THE URBAN-RURAL DIGITAL DIVIDE IN THE EUROPEAN UNION NARROWING OVER TIME?

02

- How has internet access evolved in urban and rural areas across the EU between 2006 and 2025?
- Which countries still exhibit the largest urban-rural digital gaps?
- Has convergence occurred uniformly across Member States, or do national trajectories differ?
- Are countries with initially larger gaps converging faster than early digital leaders?

About Data

*The dataset used for this analysis is "**Households – level of internet access**" from Eurostat (code: isoc_ci_in_h). It provides annual data on the percentage of households with internet access, disaggregated by degree of urbanisation.*

Dataset Selection Criteria

The following parameters were selected to ensure a consistent measurement of the **urban-rural digital divide over time**:

- **Temporal scope:** Annual data from 2006 to 2025
- **Unit of measurement:** Percentage of households (PC_HH)
- **Geographic scope:** EU countries only
- **Two disaggregation categories:** Households located in cities and households located in rural areas

Initially, the default Eurostat export did not explicitly show the urban/rural breakdown as a separate variable. The dimension "degree of urbanisation" was embedded under household type categories (e.g., "Households located in cities" and "Households located in rural areas"). This required **careful inspection of metadata and variable structure** before proceeding with the analysis.

Licence

Eurostat data are publicly available under the **Creative Commons Attribution 4.0 International (CC BY 4.0)** licence.

This allows reuse, modification and redistribution, provided attribution is given. Therefore, the data can be legally used for analysis, transformation and visualisation.

About Data

Data Quality & Limitations

Geographic Coverage

The dataset includes EU and non-EU countries (EEA, candidates, UK, etc.). For comparability, the analysis was **restricted to EU27 current members**. Country coverage varies slightly in early years — 25 countries in 2006 versus 27 in 2025.

Methodological Harmonisation

Data are based on **national surveys harmonised by Eurostat**; therefore, minor methodological changes over time may influence comparability. Percentages do not account for population distribution differences across countries.

Overall Assessment

Despite these caveats, the dataset is **highly standardised** and suitable for longitudinal comparative analysis across the EU27 member states over a nearly two-decade period.

Data Cleaning Operations

The following cleaning steps were performed in R to prepare the dataset for analysis. All transformations were documented and exported for reproducibility. The complete code is available at the following GitHub repository: **[GitHub - noemiscanu/DATA-VIZUALIZATION-PROJECT]**.

01

Rename & Convert

Renaming variables for clarity and converting year and percentage values to numeric format.

02

Filter Categories

Filtering only urban and rural categories, excluding the "total" aggregate to isolate the comparison dimensions.

03

Map & Remove

Mapping country names to ISO codes and removing non-EU countries (documented in an audit table).

04

Reshape & Compute

Reshaping data from long to wide format and creating a new variable: **Digital Gap = Urban% - Rural%**.

Methodology

Data Collection

Data were manually downloaded from Eurostat's Data Browser in CSV format, selecting relevant dimensions: **urbanisation level, EU countries, annual frequency, and percentage of households.** No scraping or APIs were used for data retrieval.

Data Cleaning

The dataset was processed in **R** using the following packages:

- `tidyverse` – core data manipulation
- `readr` – CSV import
- `janitor` – variable name cleaning
- `countrycode` – ISO code mapping
- `stringr` – string operations

Cleaning included variable renaming, filtering, recoding and reshaping operations as detailed in the data section.

Data Transformation

Several transformations were applied to prepare the analytical dataset:

- **Filtering** EU27 countries for comparability
- **Reshaping** from long to wide format
- **Computing** the digital gap (urban - rural)
- **Aggregating** yearly EU averages
- **Calculating** country-level convergence (gap reduction from first to last available year)

Analysis

The following techniques were applied:

- **Descriptive statistics:** Mean, min, max across countries and years
- **Longitudinal trend analysis:**Tracking EU averages from 2006 to 2025
- **Comparative cross-country analysis**
- **Ranking analysis** (latest year)
- **Convergence analysis:** gap reduction over time)

Use of AI Tools

ChatGPT was used to structure the analytical pipeline, refine R code, and improve clarity and logical consistency of the narrative. **All outputs were validated manually in R.**

Reporoducibility & Summary

Fully reproducible analysis based on public Eurostat data, clearly defined filtering criteria, documented R scripts, and exported outputs to ensure transparency and independent verification..

Insights from the Data

-7.4
p.p

Reduction in the average EU urban-rural digital gap (from 10.8 p.p. in 2006 to 3.38 p.p. in 2025).

+51.1
p.p

Increase in rural internet access (vs. +43.8 p.p. in urban areas), signalling faster rural growth and structural convergence.

FROM >20p.p TO <5p.p

Countries with very large gaps (e.g., Lithuania 23.6; Romania 21.9 in 2006) are now mostly clustered below 5 p.p. in 2025.

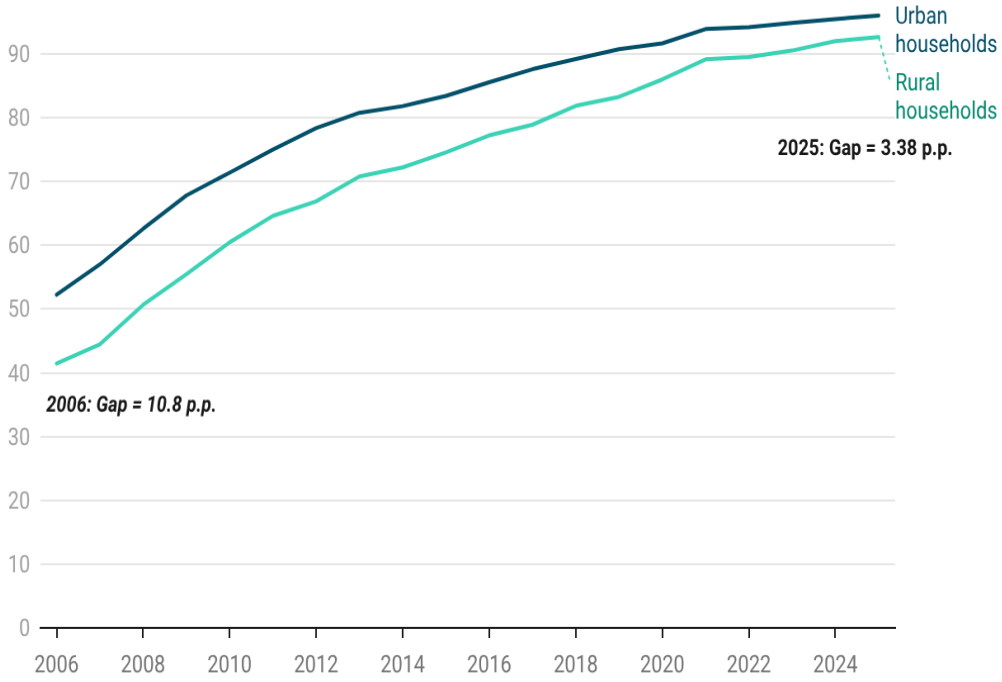
STILL >10 p.p.

A small group of countries (e.g., Greece, Portugal, Bulgaria) maintains double-digit gaps in 2025, showing uneven convergence.

DATA ANALYSIS OPERATIONS PERFORMED

- Descriptive statistics (mean, min, max, distribution)
- Longitudinal comparison (2006 vs 2025)
- Gap change calculation (Δ over time)
- Cross-country comparison

The Urban–Rural Digital Divide in the EU Has Narrowed Sharply Since 2006



EU27 annual average. Digital gap defined as percentage point difference between urban and rural household internet access.

Chart: Noemi Scanu • Source: Eurostat • Created with Datawrapper

Key findings

Over the past two decades, the European Union has experienced a substantial reduction in the urban–rural digital divide.

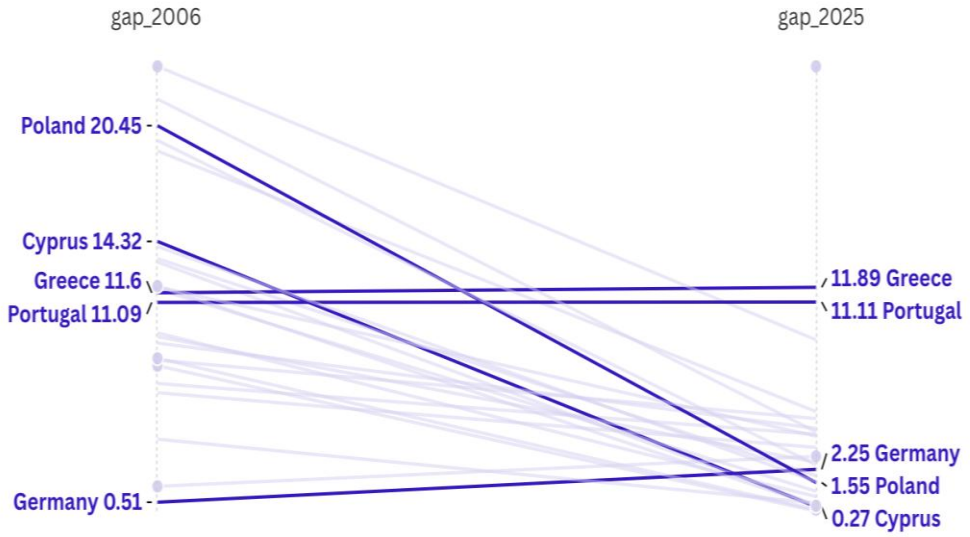
In 2006, the average gap in household internet access stood **at 10.8 percentage points**, reflecting significant territorial disparities in connectivity. By 2025, the gap narrowed to **3.4 percentage points**, corresponding to a reduction of nearly two-thirds.

This convergence was primarily driven by faster growth in rural access **(+51 p.p.)** compared to urban areas **(+44 p.p.)**, indicating structural digital catch-up.

While minor disparities persist in some Member States, the EU as a whole has reached levels of near-universal internet access.

National Trajectories Behind Europe's Digital Convergence

Urban–rural internet access gaps across EU countries show strong overall convergence, but persistent national differences in speed and direction of change.



Source: Eurostat. • Gap measured as the absolute percentage point difference between urban and rural internet access. Each line represents one country. The left axis shows the urban–rural digital gap in 2006, the right axis in 2025. Downward slopes indicate a reduction in inequality; upward slopes indicate widening gaps.

Key findings

Across Member States, the reduction of the urban-rural digital gap followed markedly different trajectories.

In 2006, several countries displayed substantial disparities, with gaps above **20 p.p** (e.g., Lithuania 23.6; Romania 21.9; Poland 20.4).

By 2025, most countries cluster below **5 p.p.**, reflecting widespread structural convergence. Poland (-18.9 p.p.), Romania (-17.6 p.p.), and Spain (-17.2 p.p.) recorded the largest improvements.

However, convergence was not universal. Greece (11.6 → 11.9 p.p.) and Portugal (~11 p.p.) show persistent gaps, while Germany and Malta experienced moderate widening.

Importantly, initial conditions did not determine long-term outcomes: Cyprus reduced its gap from **14.3 p.p.** to 0.27 p.p., whereas Germany moved from near parity (0.5 p.p.) to a wider gap (2.25 p.p.), suggesting that policy dynamics and investment trajectories played a decisive role.

Metodologiacal Note

Highlighted countries represent:(1) the largest reductions,(2) cases of stagnation or widening, and(3) a country near the 2025 median for contextual reference.

Conclusion

- Over the past two decades, Europe has moved from fragmentation toward convergence in digital access.
- Rural areas have caught up significantly, narrowing what was once a deep territorial divide.
- Yet convergence has not been uniform: while many countries approached parity, others continue to experience persistent structural gaps.
- Bridging the digital divide is no longer only about expanding infrastructure, it is about ensuring equitable and sustained inclusion across territories.

LICENCE

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